

RobustGaSP: MLE vs MAP estimation

5D test example

Friedman.5 vs my 5D function

1 Function 1:

$$y = 10 \sin(\pi x_1 x_2) + 20(x_3 - 0.5)^2 + 10x_4 + 5x_5$$

1.1 Emulator fit and validation

We have a 5-dimensional input, one output, and 50 noise-free observations. We will assume a linear mean function of the form $m(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5$. The Gauss (squared exponential) *and* Matern 5/2 kernels will be compared for both estimation methods.

100 new LH points are generated, and out-of-sample checks (as per **RobustGaSP**'s Help page) shown in Table 1, with the emulator predictions plotted against the actual outputs for these points shown in Figure 1.

Table 1: Out-of-sample checks

Estimate method	Kernel	Prop. w/n 95% CI	Out-of-sample MSE	Mean width of 95% CI
MLE	Gauss	0.57	0.01	0.1377
MLE	Matern 5/2	1	0.007	0.4197
MAP	Gauss	0.64	0.01	0.1557
MAP	Matern 5/2	1	0.005	0.4478

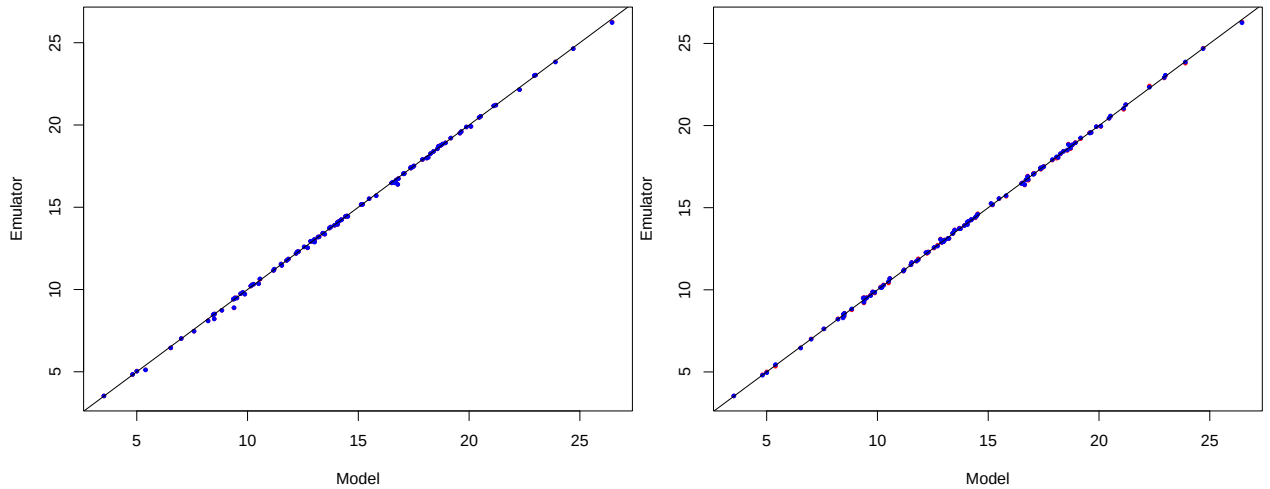


Figure 1: Emulator vs model for Gauss (left) and Matern 5/2 (right). Red: MLE. Blue: MAP

Finally, in preparation for the real world data where we cannot perform out-of-sample validation, leave-one-out validation is carried out; the results are shown in Figure 2.

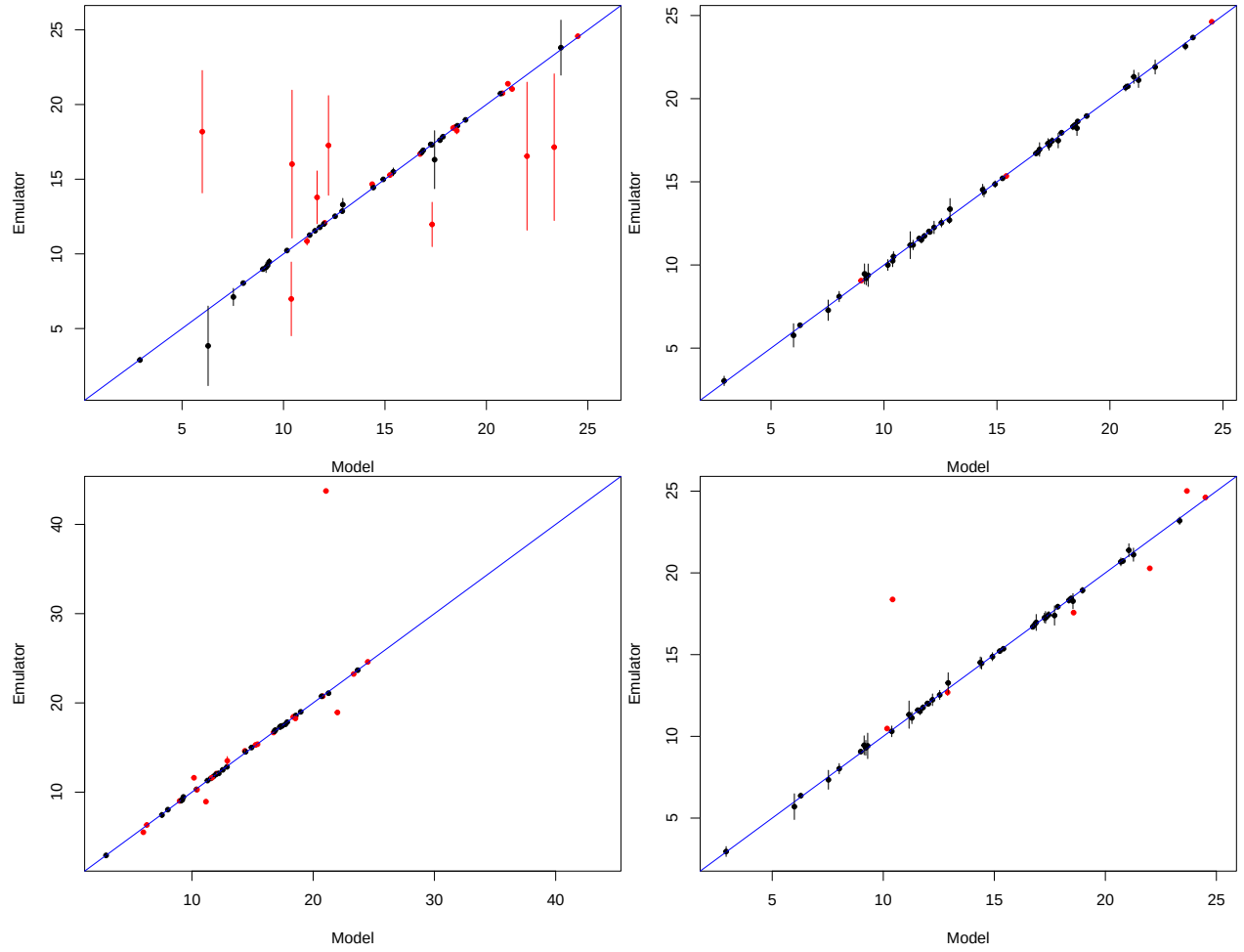


Figure 2: LOO CV. Top: MLE with Gauss (left) and Matern 5/2 (right) kernels. Bottom: MAP.

Going forward, for now, only the Gauss kernel is dealt with; Matern 5/2 will be returned to once I've figured out the estimates returned by the two packages and the derivative of the Matern 5/2 kernel.

1.2 Verifying the estimates by-hand for both estimation methods

To calculate the posterior predictions at the 100 testing points by-hand, we require the matrix H containing the inputs for the 50 observations and well as matrix A , its inverse, and $\mathbf{t}(x^*)$ for all 50 of the new inputs x^* . A and $\mathbf{t}(x^*)$ are built using the estimate of the range parameter returned by each package. By-hand predictions match those returned by the two estimation methods (Figure 3).

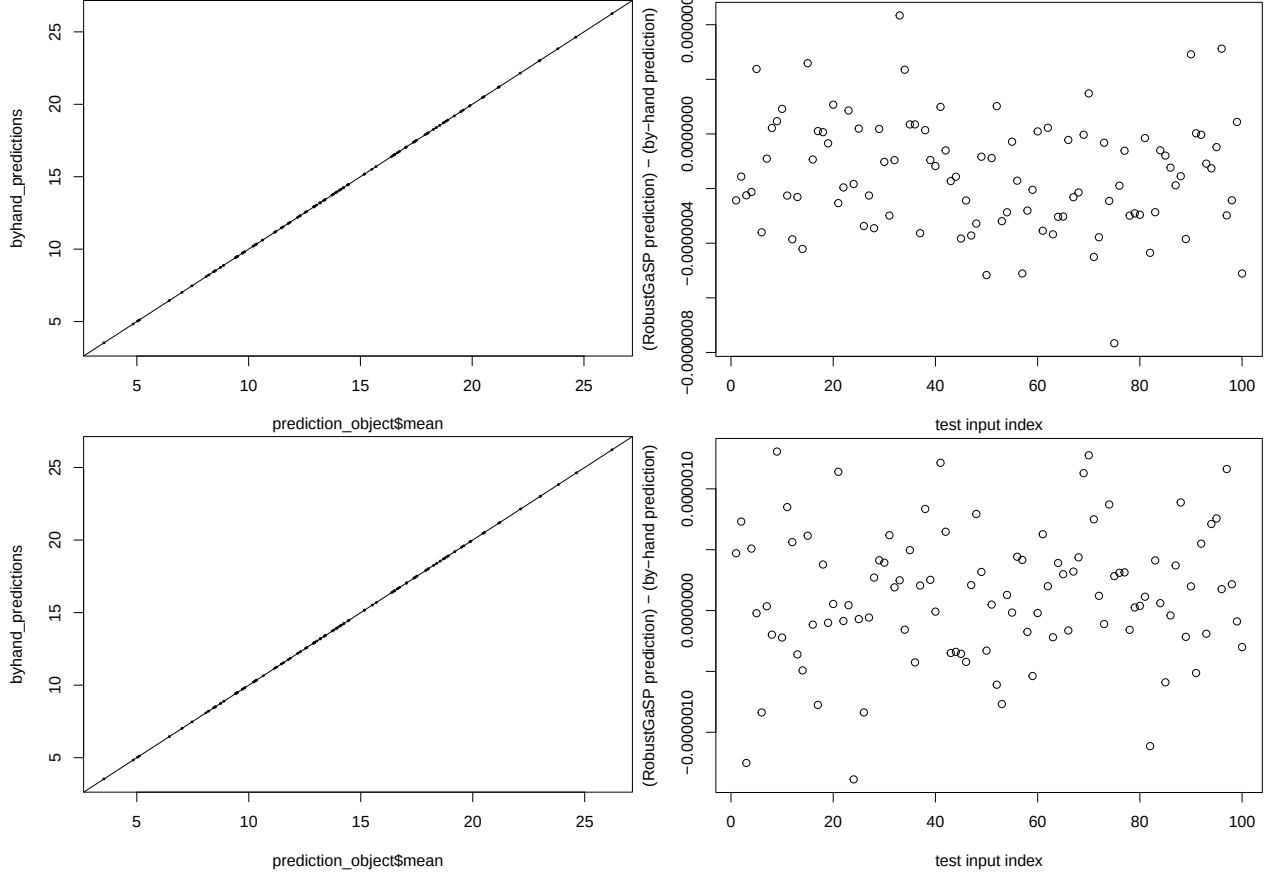


Figure 3: The predictions found by-hand match well with those made by both estimation methods (left), with the latter minus the former found to be essentially zero (right). Top: MLE. Bottom: MAP.

For both we calculate

$$\frac{\sum_{i=1}^N \left| (\text{predictions from package})_i - (\text{predictions by-hand})_i \right|}{N},$$

where N is the number of test points, then divide this by 10^{-8} (machine precision) to get 20 for MLE and 40 for MAP.

We create an object containing the emulated partial derivatives for use in the alignment measure. Finally, we create an object containing the normalised partial derivatives for use in the alignment measure.

2 Function 2:

$$y = 10 \sin(-\pi x_1 x_2) - 20(x_3 - 0.5)^2 + 10x_4 - 5x_5$$

We repeat the above steps for the above function. The same training and testing points are used as for the first function.

2.1 Emulator fit and validation

Out-of-sample check results are shown in Table 2 and Figure 4.

Table 2: Out-of-sample checks

Estimate method	Kernel	Prop. w/n 95% CI	Out-of-sample MSE	Mean width of 95% CI
Maximum likelihood	Gauss	0.57	0.01	0.1377
Maximum likelihood	Matern 5/2	1	0.007	0.4197
MAP	Gauss	0.64	0.01	0.1557
MAP	Matern 5/2	1	0.005	0.4478

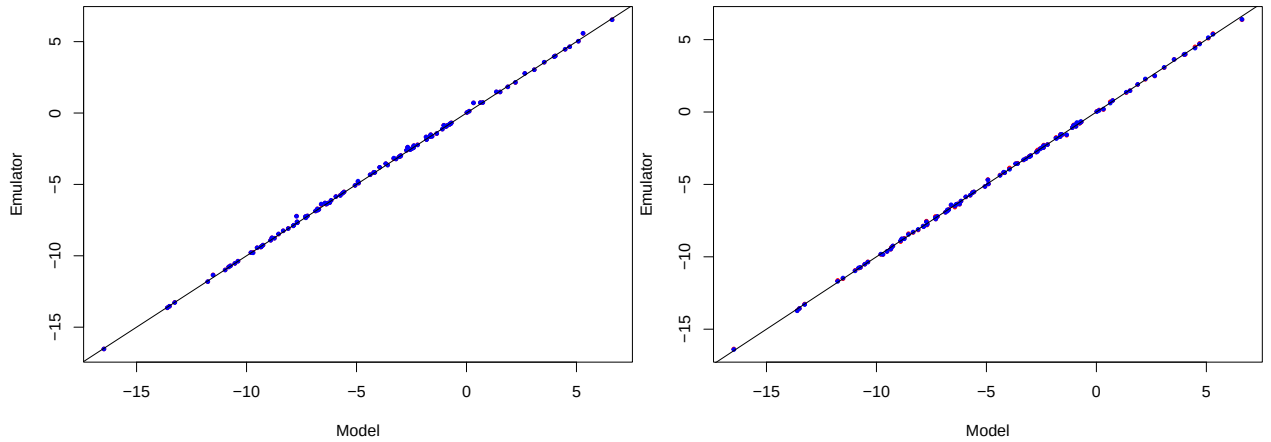


Figure 4: Emulator versus model for the new points. Left: Using a Gauss kernel. Right: Using a Matern 5/2 kernel. Red: MLE. Blue: MAP

LOO CV results are shown in Figure 5, but only for the Gauss kernel as some of the LOO predictions for the Matern 5/2 kernel (in both estimation methods), extremely large positive or negative values were returned (along with NaN for the corresponding lower or upper bound of the 95% CI).

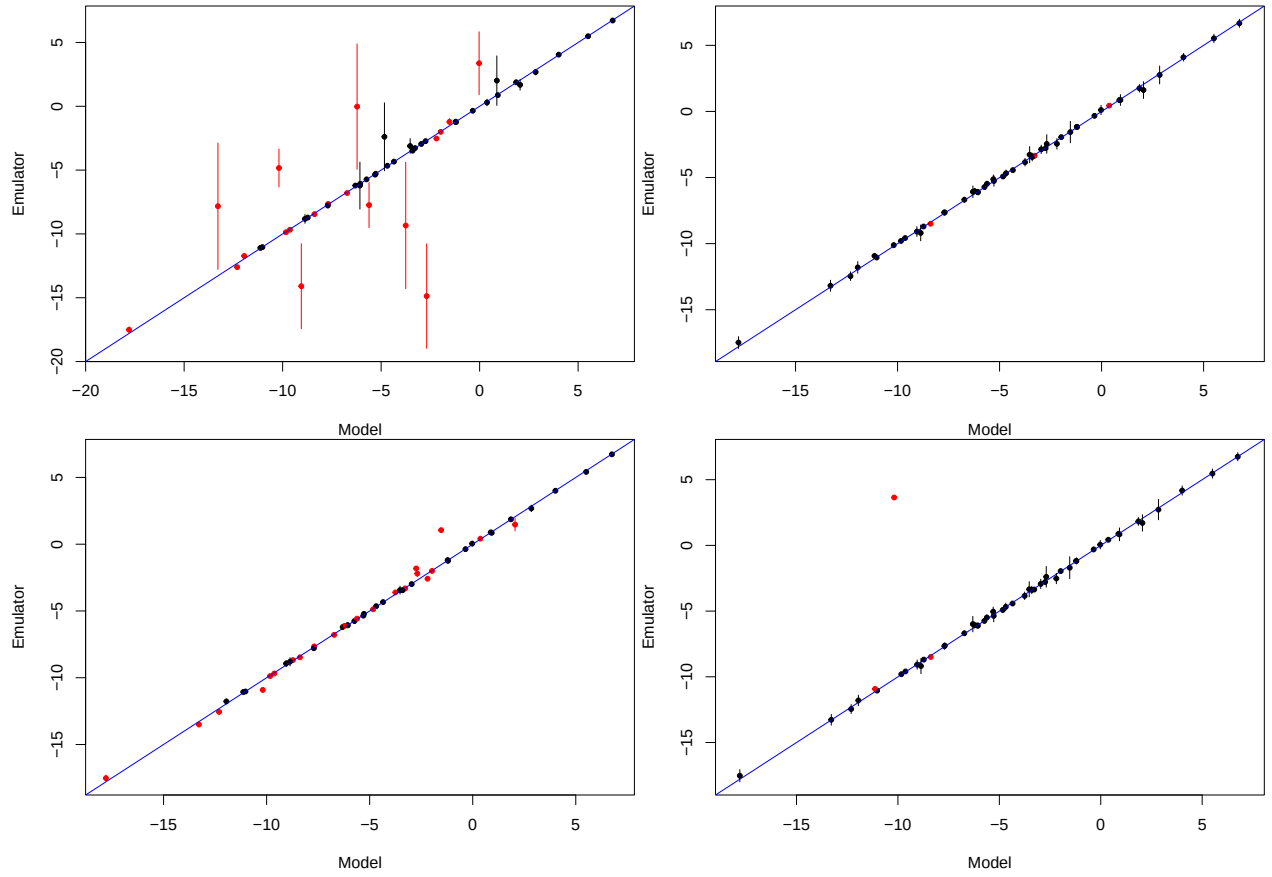


Figure 5: LOO CV. Top: Maximum likelihood estimation for the Gauss (left) and Matern 5/2 (right) kernels. Bottom: Marginal posterior mode estimation for the Gauss (left) and Matern 5/2 (right) kernels.

2.2 Verifying the estimates by-hand for both estimation methods

A good match is found for the predictions (Figure 6).

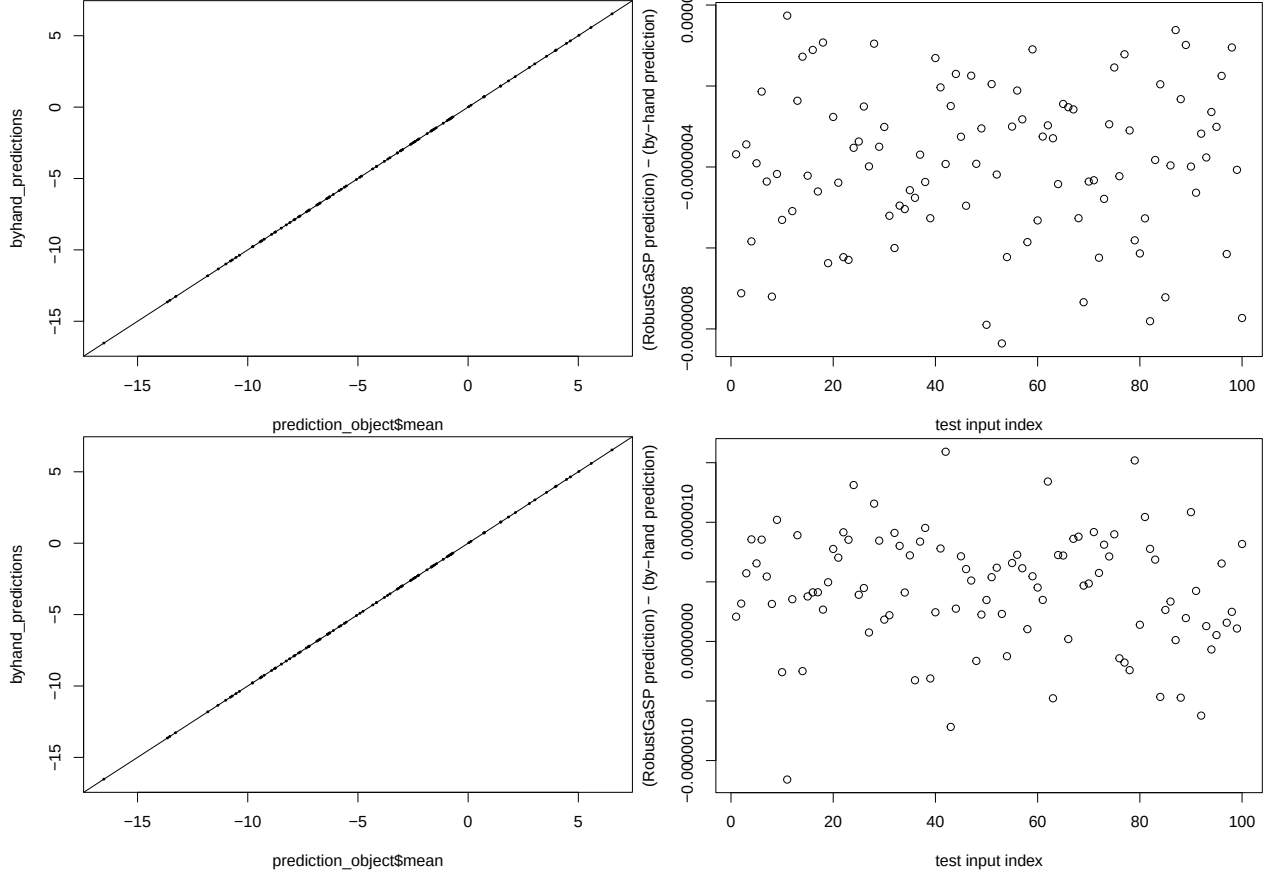


Figure 6: The predictions found by-hand match well with those made by both estimation methods (left), with the latter minus the former found to be essentially zero (right). Top: MLE. Bottom: MAP.

For both we calculate

$$\frac{\sum_{i=1}^N \left| (\text{predictions from package})_i - (\text{predictions by-hand})_i \right|}{N},$$

where N is the number of test points, then divide this by 10^{-8} (machine precision) to get 40 for MLE and 60 for MAP.

3 Alignment measure

We calculate the alignment measure of this function with a couple of others. Using the **actual** partial derivatives of the two functions at the 100 testing points, we find an AM of 0.5638 (and 0.5591 using 1000). Using MLE at these 100 points we get an alignment for these two functions of 0.5662 (and 0.5574 using 1000) and for MAP 0.5663 (and 0.5575 using 1000).